

Neural Sheaf Diffusion: A Topological Perspective on Heterophily and Oversmoothing in GNNs

Summary

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1 Problem Setup

Graph Neural Networks (GNNs) struggle with two fundamental problems: oversmoothing (where node representations become identical after many layers) and heterophily (where connected nodes tend to have different labels). Traditional GNNs assume homophily (connected nodes have similar labels) and use simple message passing that leads to oversmoothing. This paper introduces sheaf theory to address both issues simultaneously.

2 Thesis

The central thesis is that sheaf theory provides a natural mathematical framework for GNNs that can handle heterophily and prevent oversmoothing. By learning sheaf restriction maps that transform features between connected nodes, the model can adaptively align or separate representations based on the relationship between nodes.

3 Key Concepts

Cellular Sheaves: Mathematical objects that assign vector spaces (called "stalks") to nodes and linear maps (called "restriction maps") to edges of a graph. These maps determine how features transform between connected nodes.

Sheaf Laplacian: A generalization of the graph Laplacian that incorporates the restriction maps. It governs how information diffuses across the graph according to the learned transformations.

Sheaf Diffusion: The process of repeatedly applying the sheaf Laplacian to update node representations. Unlike standard diffusion, sheaf diffusion can rotate, scale, or otherwise transform features as they propagate.

Heterophily vs Homophily: Homophily means connected nodes have similar labels; heterophily means they have different labels. Sheaf diffusion can handle both by learning appropriate restriction maps.

4 Methods

The authors develop Neural Sheaf Diffusion (NSD) models with several variants:

- **Diagonal Sheaves:** Restriction maps are diagonal matrices, allowing simple scaling of features.

- **Orthogonal Sheaves:** Restriction maps are orthogonal matrices, allowing rotations but preserving distances.
- **General Sheaves:** Full linear transformations between node features.

The learning process involves:

1. Learning restriction maps from data using neural networks
2. Constructing the sheaf Laplacian from these maps
3. Performing diffusion using the Laplacian
4. Using the diffused representations for downstream tasks (typically node classification)

The method can be implemented in both discrete (layer-by-layer) and continuous (differential equation) forms.

5 Results

The authors demonstrate superior performance on node classification benchmarks, especially on graphs with low homophily:

- **Heterophilic Graphs:** NSD significantly outperforms standard GNNs on graphs where connected nodes have different labels.
- **Homophilic Graphs:** NSD matches or slightly improves upon standard GNN performance.
- **Oversmoothing Resistance:** NSD maintains distinct node representations even after many diffusion steps.
- **Learned Transformations:** The model learns meaningful restriction maps that rotate or scale features based on node relationships.

Theoretical results show that sheaf diffusion can perfectly solve certain classification tasks that are impossible for standard GNNs.

6 Significance

This work bridges sheaf theory and deep learning, providing:

- **Mathematical Foundation:** A principled approach to GNN design based on topological data analysis.
- **Practical Benefits:** Improved performance on challenging graph learning tasks, especially with heterophily.
- **New Research Direction:** Opens up connections between algebraic topology and graph representation learning.
- **Flexibility:** The framework can incorporate various types of learned transformations while maintaining theoretical guarantees.

The paper demonstrates that abstract mathematical concepts from sheaf theory can lead to practical improvements in machine learning, particularly for graph-structured data.